

NETFLIX DATA ANALYSIS

Using Python, Pandas, Matplotlib & Seaborn

INTERNSHIP PROJECT REPORT

Auspify Technologies - Data Analysis Using Python Internship Program

Author	Jogi Dhilleswar
Completion	6/6 Tasks Completed
Environment	Python / Jupyter Notebook / VS Code
GitHub	https://github.com/dhilleswar07/Netflix_Data_Analysis

This report documents the six-task Netflix analysis workflow, analytical objectives, outputs, tools, skills demonstrated, and business-analysis scope.

Executive Summary

This project is an end-to-end Netflix Data Analysis project completed as part of the Auspify Technologies Data Analysis Using Python Internship Program. The project covers data cleaning, content-type analysis, country-wise analysis, release-year trends, rating and genre analysis, and a final business insights report. The work is organized into task-specific Jupyter notebooks with CSV outputs, following a reproducible workflow from raw data through cleaning, exploratory analysis, visualization, and business-oriented reporting.

Completion: All six tasks have been completed. The internship task list states that interns need to complete any four of the six tasks; this project covers the complete six-task scope

Project Objectives

- Prepare and clean the Netflix dataset for reliable analysis.
- Analyse the distribution of Movies and TV Shows.
- Study Netflix content distribution across countries.
- Analyse content release trends by year.
- Explore ratings and genres and compare content types.
- Produce business-oriented insights and analytical outputs.
- Organize notebooks and CSV outputs for professional submission.

Dataset & Project Structure

The project uses a Netflix titles dataset with fields including show ID, type, title, director, cast, country, date added, release year, rating, duration, listed genres, and description. The raw dataset is retained as Dataset.csv. The cleaned dataset is exported as Netflix_Cleaned.csv and used as the basis for subsequent analysis.

Tools & Technologies

Technology	Purpose
Python	Core programming and analytical workflow
Pandas	Data loading, cleaning, transformation and aggregation
Numpy	Numerical processing
Matplotlib	Charts and trend visualizations
Seaborn	Statistical and comparative visualizations
Jupyter Notebook	Interactive task-by-task analysis

VS Code	Development environment
GitHub	Version control and project presentation

Task 1 — Netflix Data Cleaning & Preparation

Objective: Prepare and organize the Netflix dataset for business analytics and reporting.

- Import the dataset using Python and Pandas.
- Identify and handle missing values.
- Remove duplicate records and formatting inconsistencies.
- Standardize Country, Rating, and Type.
- Export the cleaned dataset.

Output: Netflix_Cleaned.csv

Skills: Panda's fundamentals, data cleaning, data preprocessing, data quality management.

Task 2 — Content Type Analysis Dashboard

Objective: Analyse the distribution of Movies and TV Shows available on Netflix.

Content Type	Count	Share
Movies	6126	69.69%
TV Shows	2664	30.31%
Total	8790	100%

The completed analysis shows that Movies form the larger share of the dataset: 6,126 Movies (69.69%) compared with 2,664 TV Shows (30.31%).

Skills: Data visualization, Matplotlib, Seaborn, exploratory data analysis

Task 3 — Country-Wise Netflix Content Analysis

Objective: Analyse Netflix content availability across different countries.

- Extract and clean country information.
- Calculate content count by country.
- Identify top content-producing countries.
- Create charts and rankings.

- Generate business insights.

Output: Netflix_Country_Analysis.csv

Skills: Group By operations, data aggregation, business analytics, visualization.

Task 4 — Trend Analysis by Release Year

Objective: Analyse how Netflix content production has changed over time.

- Organize content by release year.
- Calculate yearly content releases.
- Identify growth and decline trends.
- Create trend visualizations.
- Interpret business patterns.

Output: Netflix_Year_Analysis.csv

Skills: Time-series analysis, trend identification, data interpretation, reporting.

Task 5 — Content Rating & Genre Analysis

Objective: Examine content ratings and genres to understand audience targeting.

- Analyse rating categories.
- Identify common genres.
- Compare ratings across Movies and TV Shows.
- Create rating and genre visualizations.
- Present audience insights.

Outputs: Netflix_Rating_Analysis.csv; Netflix_Rating_Type_Analysis.csv;
Netflix_Genre_Analysis.csv

Skills: Advanced data analysis, feature exploration, visualization design, insight generation.

Task 6 — Netflix Business Insights Report

Objective: Create a comprehensive business intelligence report from the Netflix dataset.

- Perform complete exploratory data analysis.
- Identify key content trends and patterns.

- Develop multiple visual dashboards.
- Generate actionable business insights.
- Prepare and present the final analytical report.

Output: Netflix_Business_Insights_Report.csv

Skills: End-to-end data analysis, business analytics, dashboard development, data storytelling, data-driven decision making

Key Analytical Findings

- Movies represent the majority of the analysed title records.
- Country-wise analysis provides visibility into the geographic concentration of Netflix content.
- Release-year analysis provides visibility into historical catalog and production trends.
- Rating analysis supports comparison of audience-oriented content categories.
- Genre analysis helps identify the types of content most represented in the catalog.
- Combining content type, country, year, rating, and genre creates a broader business intelligence view.

Project Outputs

File	Purpose
Dataset.csv	Raw dataset
Netflix_Cleaned.csv	Cleaned dataset
Netflix_Country_Analysis.csv	Country analysis
Netflix_Genre_Analysis.csv	Genre analysis
Netflix_Rating_Analysis.csv	Rating analysis
Netflix_Rating_Type_Analysis.csv	Rating comparison by content type
Netflix_Year_Analysis.csv	Release-year analysis
Netflix_Business_Insights_Report.csv	Final business insights output

Skills Demonstrated

- Python programming
- Pandas and NumPy

- Data cleaning and preprocessing
- Exploratory data analysis
- Matplotlib and Seaborn visualization
- GroupBy and data aggregation
- Time-series and trend analysis
- Business analytics and data storytelling
- Dashboard-oriented reporting
- Insight generation and professional project organization

Conclusion

The Netflix Data Analysis project demonstrates a complete practical data-analysis workflow, beginning with raw data preparation and progressing through descriptive analysis, visualization, trend analysis, rating and genre exploration, geographic analysis, and business-oriented reporting. The six notebooks and supporting CSV outputs make the work reproducible and suitable for internship evaluation and portfolio presentation.

Final Submission Checklist

- Include all completed tasks and supporting files.
- Include source notebooks, reports, screenshots, and documentation.
- Compress the relevant project files into a single ZIP file.
- Exclude unnecessary files such as virtual environments and cache folders.
- Verify that the GitHub repository and other submitted links are accessible.
- Use a clear project folder structure for evaluation.

Links

GitHub Repository: https://github.com/dhilleswar07/Netflix_Data_Analysis

Portfolio: <https://dhilleswar07.github.io/>

LinkedIn: <https://www.linkedin.com/in/dhilleswar07/>